

Modelling and Control of planar Soft Robot using Mass Spring System

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Dual Degree (B. Tech. + M. Tech.)

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Abstract

Despite the recent advancements in the field of Soft Robotics, developing model-based control for the same remains a challenge. This is largely due to the lack of real-time models to predict its behavior and the infinite dimensions of the soft robot. In this paper, we propose the use of a novel Mass Spring System to model soft robots under planar loading. The effectiveness of this model is evaluated by comparing the static deformation with FEM and analytical results. The main aim of this work lies around the development of a control strategy for soft robots using the model. The developed control strategy is used to control a 5-segment continuum robot. The experimental results of the same showed an accuracy of 0.4mm in tip-pose movement and an average error of about 1mm in trajectory tracking.

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Contents

Contents	v
List of Figures	vi
List of Tables	viii
1 Introduction	1
2 Model	3
2.1 Model description	3
2.2 Simulating the Model	4
3 Bench-marking the Model	7
4 Control Design	11
4.1 controller design	11
5 Simulation Results	14
6 Hardware Design and Fabrication	16
6.1 Fabrication of single segment	16
6.2 Torque sensor Design	17
6.3 Assembly of full robot	19
6.4 Sensing	19
6.5 Full Control Architecture	20
7 Experimental Results	24
7.1 Slope control Results	24
7.2 Set-point Control	24
7.3 Trajectory tracking	24
8 Discussions and conclusions	28
Bibliography	29

List of Figures

2.1	Representation of different robot geometry with square mesh. A beam, a rectangular plate with holes, and a circular plate each have been shown in the meshed form. The resolution or number of squares can be increased to better represent circular geometry.	3
2.2	The mass-spring network for a beam is represented. For squares sharing the same edges, the springs are added in parallel. The numbered vertices indicate the nodes and their id.	4
2.3	Deformation of the beam under distributed bending load with different mesh resolution and its comparison with corresponding FEM result. . .	5
3.1	Bending of the beam under pure vertical bending load such that the tip slope is at a desired value. The result from MSS and CBCM models are compared.	8
3.2	Bending of the beam under pure moment bending torque such that the tip slope is at a desired value. The result from MSS and CBCM models are compared.	8
3.3	Bending of the beam under uniformly distributed bending load. The predictions of the proposed model are compared with FEM results. . .	9
3.4	The dynamic response of the FEM model compared with the model predictions of a beam under step force applied at the tip at $t = 0sec$. . .	10
4.1	Continuum robot design	11
4.2	ROS framework of the implemented system for simulating the real system and running model predictions in parallel for control action decision along with information flow is pictured.	13
5.1	going to a point	14
5.2	Following an elliptical path with the minor axis of the ellipse along the center-line at the initial configuration of the robot.	15
6.1	Design of the 5 segment 2D continuum robot	16
6.2	One soft segment with the central core made of silicone polymer. The blue parts are 3D printed with ABS and provide the necessary rigidity to apply a distributed torque at the tip of the segment. The strings connected to the servo through the pulley act as an actuator.	17

6.3	The images on the left show the mold, where the rigid attachments and string guide have been put in the mold. After pouring and curing the mold is removed and the extra material on the top is cut.	18
6.4	The mold for the cylindrical torque sensor is 3D printed in 2 parts. On either side, a rigid part is put in to transfer the load and act as support for bolting it to other parts.	18
6.5	For sensor characterization, a mass of known weights is hung from the string, and the deformation is recorded through the encoder while the servo is commanded to hold the position.	19
6.6	The torque applied on the segment is transferred through the torque sensor. The slope of the segment for any given torque is captured through the camera.	20
6.7	Left images shows the complete assembly of the 5-segment continuum robot with markers placed on each segment for tracking. The top right image depicts the robot being placed on a horizontal frame with the camera mounted on the top, while the bottom right image depicts the 5 servos being used for actuating the segments.	21
6.8	Box plot of measurement of pose and slope at a given position is shown in the above plots.	22
6.9	variation of required torque with increasing slope predicted from the model is shown here.	22
6.10	The control diagram of the modified controller after implementing slope control at a low level as per hardware requirement.	23
6.11	The flow diagram explaining the hardware architecture of the experimental setup	23
7.1	Slope versus time plot obtained while tracking absolute slope as specified by the red dotted line for each segment.	25
7.2	The path and time taken to move from zero-config to various set-point using the proposed controller is highlighted.	26
7.3	The performance of the controller while following different trajectories in the workspace is depicted.	27

List of Tables

2.1	Convergence of the solution with increasing mesh resolution	6
3.1	Tensile and Compressive load results	7
3.2	Natural Frequencies Comparison	9

Chapter 1

Introduction

In recent years, research on toxic waste sampling, nuclear equipment [1,2], biomedical devices [3], pipeline monitoring [4,5] etc has been on focus. When it comes to such narrow, complicated, or sensitive task spaces soft flexible robots are preferred over rigid link robots due to their maneuverability, slim body, and flexible design [6]. Many scholars and researchers have focused their attention on developing control and design strategies for soft robots. Soft Robots have an excellent ability to bend, avoid obstacles and change shape to adapt to the environments. In addition, it is safe to work with, as the soft and flexible design ensures the least damage in case of mishaps. A lot of progress has been made in developing actuators [8,9], sensors [11–13] and platform [10] for soft robots. However due to redundancy in design and infinite dimensional workspace owing to continuously deformable structure, the development of a model-based control strategy becomes challenging under strong under actuation. More importantly, the development of a model to predict soft robot kinematics and dynamics in real-time is a challenge and the existing models predict the behavior of only a certain type of soft robot within certain assumptions [14–16].

At the current level of state of the art, developing a control strategy for infinite dimensional space is challenging. As a result, much of the emphasis has been on developing machine learning-based models for controlling and modeling soft robots [17,19]. This approach is limited to the availability of data or collecting data from experiments. In either case, iterations become a time taking and expensive process. Moreover, these models fail to give insights into the actual dynamics and kinematics. The model-based strategies are limited to quasi-static kinematics model based control [20]. However, such control strategies are limited to slow trajectories and cannot be used to trace a smooth path. Nevertheless, the focus is shifting to model-based control with the development of a finite-dimensional model to predict the behavior within reasonable accuracy [21,22].

The first step towards developing model-based control for soft robots was using a reduced order model to convert the system into a fully actuated one and then developing a control strategy for the same [23]. Many of these control strategies are based on approximate models with the control action acting directly at the state to be controlled [24]. This simplifying assumption on the model leads to lower controller performance in terms of task-space control. To the best of our knowledge, only a couple of work presents a control strategy that solves the problem of controlling the

finite-dimensional under-actuated soft robot. [25] Models a soft pendulum using affine curvature and then proposes an under-actuated controller for stabilizing the pendulum in an upright unstable configuration. There are some other works that use the Cosserat rod theory and PDE-based models for developing a control strategy, but the control accuracy in even for task-space control has been limited.

The contribution of this work towards modeling and controlling soft robots are:

- Validation of MSS model [7] in large deformation range and its use for modeling soft robot
- Developed a Simulation Package: MATLAB GUI for simulation configuration file preparation and ROS package for MSS simulation and data processing and visualization in MATLAB
- A control architecture for tip pose control of soft robot using MSS model
- Design and fabrication of a 5 segment soft continuum robot with actuation
- Design, fabrication and testing of a in house torque sensor made of silicone material
- Implementation of vision-based sensing for slopes and position with a resolution of 0.005rad & 0.2mm
- Experimental validation of control strategy on the continuum robot achieving an accuracy of less than 0.2mm in tip positioning and less than 0.8mm average error in trajectory tracking

The remainder of this paper is organized as follows. Section 2 introduces the extension of the Mass Spring Model [7] to model the kinematics and dynamics of the soft robot. In section 3 we show the effectiveness of the model by comparing its predictions with existing theories and FEM. In section 4 we propose a control strategy for controlling continuum robots which can be generalized to many other soft robots. The simulation results of the controller are presented in section 5. Section 6 presents the hardware design and explains the experimental setup. The experimental results are discussed in section 7. Finally, the discussion and conclusion are presented in section 8.

Chapter 2

Model

In this section, we propose a framework for modeling the kinematics and dynamics of the soft robot using a Mass Spring System. The proposed work is limited to the loading conditions of plane stress. The deformation along the width of the robot is assumed to be the same. Under such assumptions, the robot design can be represented by a square mesh in 2D.

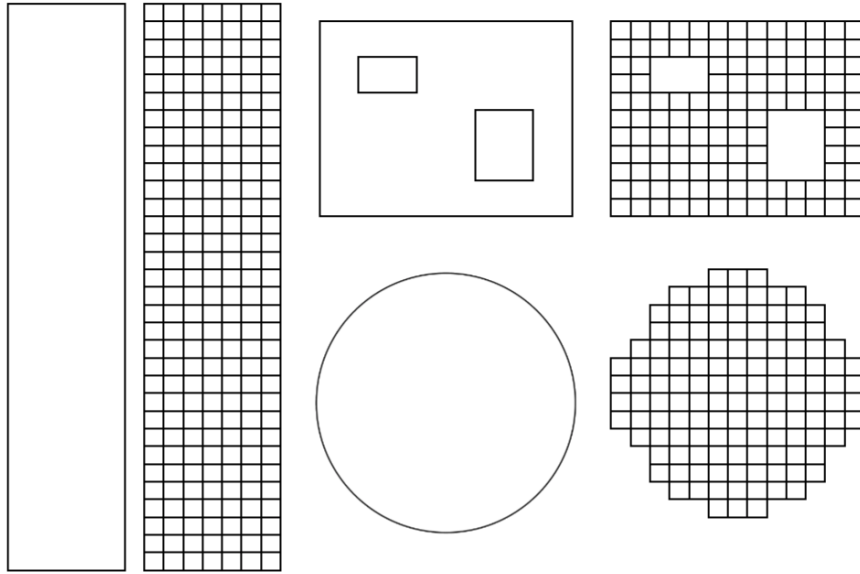


Figure 2.1: Representation of different robot geometry with square mesh. A beam, a rectangular plate with holes, and a circular plate each have been shown in the meshed form. The resolution or number of squares can be increased to better represent circular geometry.

2.1 Model description

The robot design is represented as a combination of square elements as in Fig. 2.1. Each vertex is considered a node and squares are assumed to have the Mass Spring Topology as proposed in [7]. The density of the material is used to find the mass of every square element which is then distributed equally among the 4 vertices to find the mass of each node. The dynamics of the robot can then be represented by

$$M\ddot{X}(t) + K(X)X(t) = F(t)$$

$$X = \begin{bmatrix} x1 \\ y1 \\ \cdot \\ \cdot \\ \cdot \\ xn \\ yn \end{bmatrix} \quad M = \begin{bmatrix} m_1 & & & & & \\ & m_1 & & & & \\ & & \cdot & & & \\ & & & \cdot & & \\ & & & & \cdot & \\ & & & & & m_n \\ & & & & & & m_n \end{bmatrix} \quad (2.1)$$

where x'_i 's and y'_i 's are the x and y displacement of each node from the rest positions, m'_i 's are the masses of the nodes and $K(X)$ is the stiffness matrix for the network shown in Fig 2.2. Note, as the shape of the robot changes with time, the stiffness matrix will also change. This is the key feature of this model leading to the capture of geometric non-linearity.

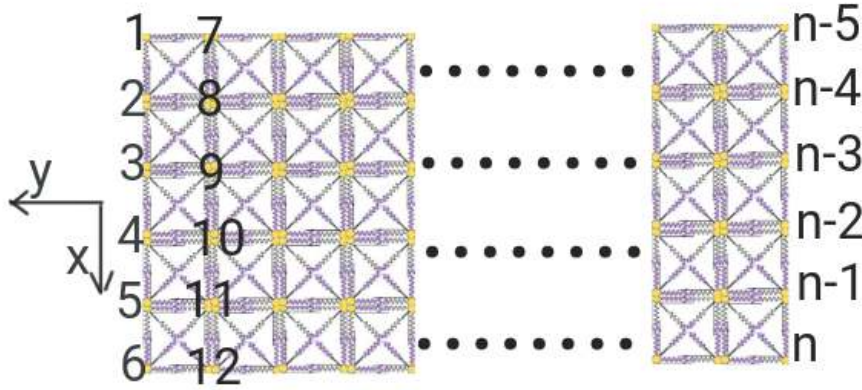


Figure 2.2: The mass-spring network for a beam is represented. For squares sharing the same edges, the springs are added in parallel. The numbered vertices indicate the nodes and their id.

To simulate the effect of damping, Rayleigh Damping is added to the equation (2.1) resulting in

$$M\ddot{X}(t) + (\alpha M + \beta K(X))\dot{X}(t) + K(X)X(t) = F(t) \quad (2.2)$$

where α and β are the mass damping coefficient and stiffness damping coefficient respectively.

2.2 Simulating the Model

To evaluate $X(t)$ in equation (2.2) under the effect of external load $F(t)$, equation 2.2 is discretized in time as given by

$$M\ddot{X}_i + (\alpha M + \beta K_i)\dot{X}_i + K_i X_i = F_i \quad (2.3)$$

where X_i represents the state $X(T_i)$, T is the time-step size and K_i is the spring stiffness of the network shown in Fig 2.2 in configuration X_i .

Implicit Newmark Beta method is used to evaluate equation (2.3) as mentioned in [27] with Newmark $\gamma = 0.5$ and $\beta = 0.25$. To efficiently perform all the above computations open source simulator VegaFEM [26] was used with appropriate modifications. Note, the second-order dynamics integrator, Implicit Newmark Beta gives stable results of the equations as far as the time-step is small enough to capture all the system frequencies. Any joint constraints on the model can be taken care of by setting the deformation of the corresponding node to zero and removing that row from the equation (2.3).

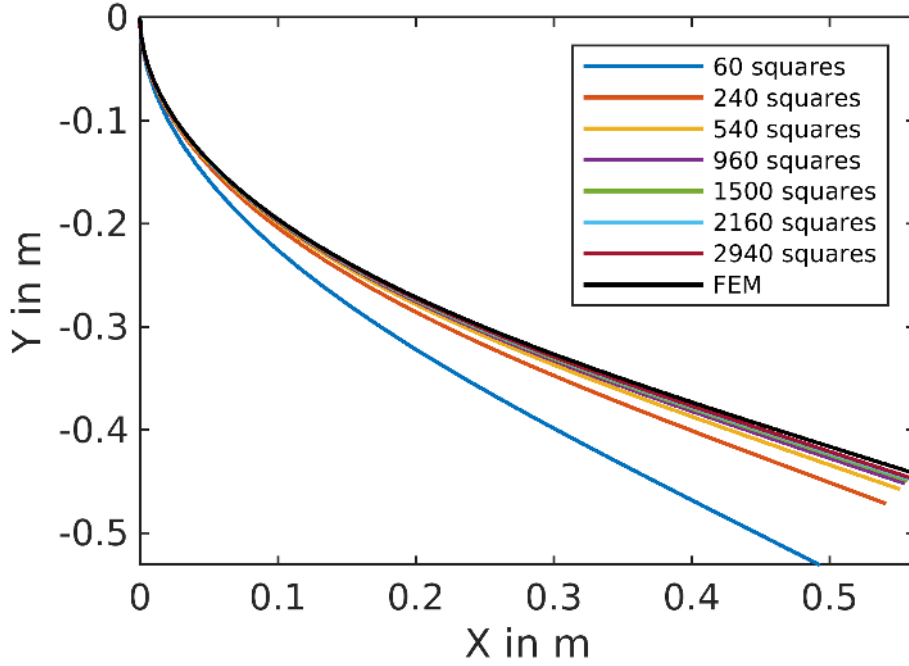


Figure 2.3: Deformation of the beam under distributed bending load with different mesh resolution and its comparison with corresponding FEM result.

A beam of the same shape and size can be represented with varied mesh resolution or the number of squares. This has a direct impact on the accuracy of the model when compared to FEM. Fig 2.3 and table 2.1 shows the bending of $0.05m \times 0.05m \times 0.75m$ cantilever beam with elasticity $1MPa$ under a uniformly distributed bending load of $10N$. As the resolution of the mesh increases the predictions from the MSS model move closer to the actual results and eventually converge to FEM results within 0.55% error. The last column of the table. 2.1 indicated the time taken by MSS for simulating 9s of the beam dynamics. The computation time for the model to predict the behavior within 4% error is 1/30th of the simulation time which is quick enough for using this model for control purposes.

Table 2.1: Convergence of the solution with increasing mesh resolution

Number of squares	Tip Deflection MSS	Tip Deflection FEM	%error	Simulation time in <i>ms</i>
60	0.492879	0.5619	12.283	53
135	0.527908	0.5619	6.04947	144
240	0.541015	0.5619	3.7168	306
375	0.547389	0.5619	2.5824	554
540	0.550998	0.5619	1.9402	908
735	0.553277	0.5619	1.5346	1384
960	0.554807	0.5619	1.2623	2015
1215	0.5559	0.5619	1.0678	2763
1500	0.556707	0.5619	0.9241	3721
1815	0.557324	0.5619	0.8143	4867
2160	0.55782	0.5619	0.7261	6115
2535	0.55882	0.5619	0.5481	7788
2940	0.558304	0.5619	0.6399	9642
3375	0.558802	0.5619	0.55135	12077
3840	0.559029	0.5619	0.5109	14435

Chapter 3

Bench-marking the Model

The effectiveness of the model in simulating soft robots was studied by performing simulations on beams under different loading conditions. At first, a soft beam $0.05m \times 0.05m \times 0.75m$ with Young's modulus $1 \times 10^6 Pa$ is loaded under tensile and compressive force with roller joint constraint at the other end of the beam. It can be observed from table 3.1 that in a small deformation regime, the results match well with analytical results while in the large deformation regime, the results start deviating as the model captures geometric non-linear behavior while the analytical method is linear.

Table 3.1: Tensile and Compressive load results

Force(N)	Deformation MSS	Deformation analytical	% Error
1000	0.2689354	0.3	10.4
500	0.1410897	0.15	5.94
80	0.0237233	0.0240	1.15
10	0.00299275	0.003	0.242
-10	-0.00300076	-0.003	0.0253

The bending of the beam under pure vertical end force and the pure moment was simulated. For this, a beam $0.1m \times 0.1m \times 1m$ with Young's modulus $1.4 \times 10^9 Pa$ with one end fixed was subjected to pure vertical force. The results obtained here are compared with Chained Beam Constrained Model (CBCM) [28] as shown in Fig. 3.1 and Fig. 3.2. The model prediction is in good agreement with CBCM and the beam profile matches reasonably well at all points along the length of the beam under both loading conditions.

The model also performs well under distributed load. Deformation of beams of Young's modulus $1MPa$ with rectangular cross-section of $0.05m \times 0.05m$ with varying length-to-width ratio was simulated under a uniformly distributed bending load of 10N for each beam. The obtained result was compared with FEM results as shown in Fig. 3.3. The error in tip deflection of MSS prediction with FEM is 2.8%, 1.7% and 0.7% respectively for L/B ratios of 6, 10, and 15. The error for an even smaller L/B ratio is less than 3% which is not the case in most beam-based theories

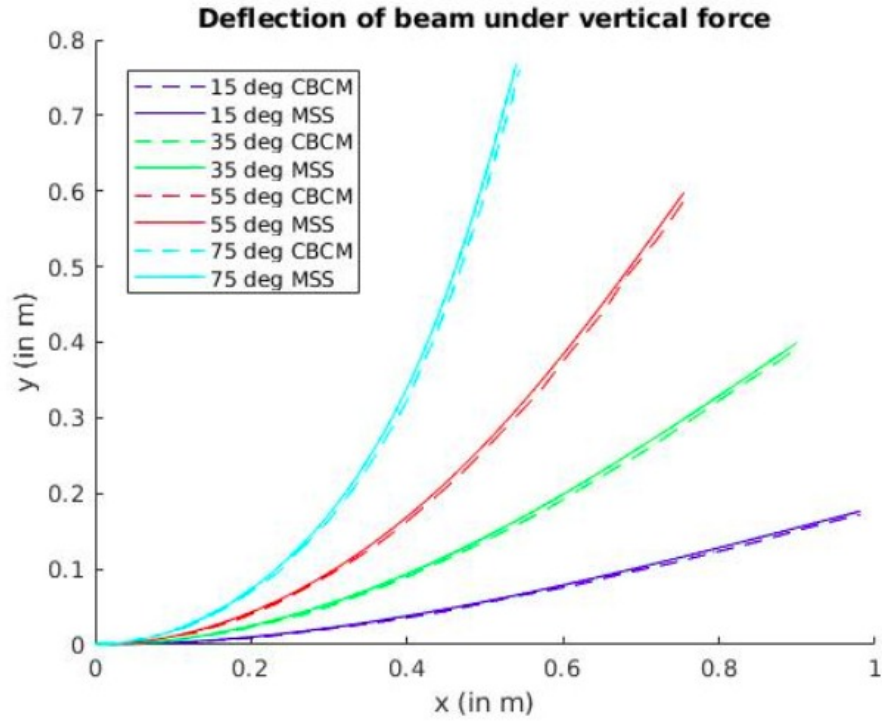


Figure 3.1: Bending of the beam under pure vertical bending load such that the tip slope is at a desired value. The result from MSS and CBCM models are compared.

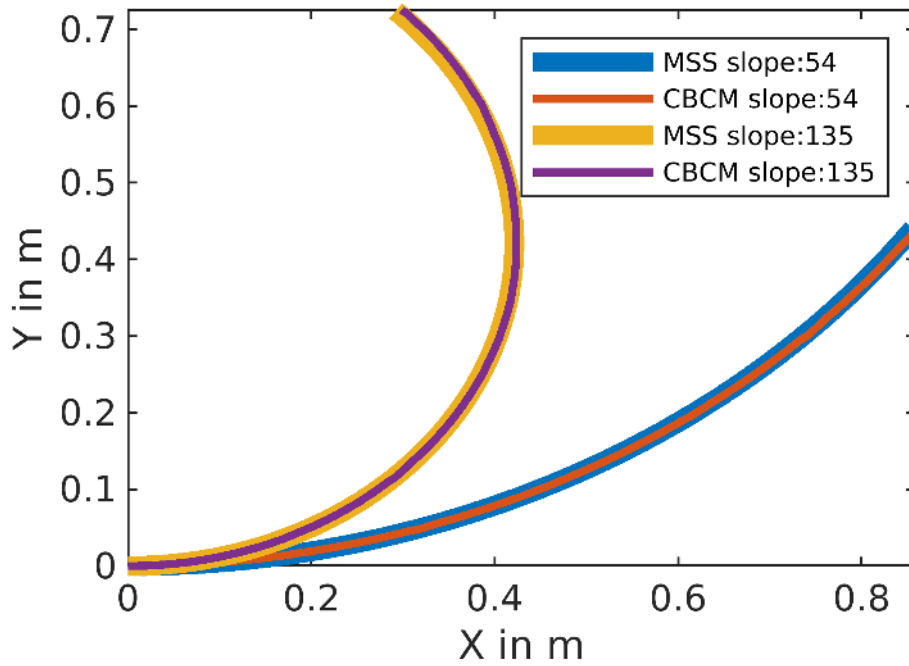


Figure 3.2: Bending of the beam under pure moment bending torque such that the tip slope is at a desired value. The result from MSS and CBCM models are compared.

or models.

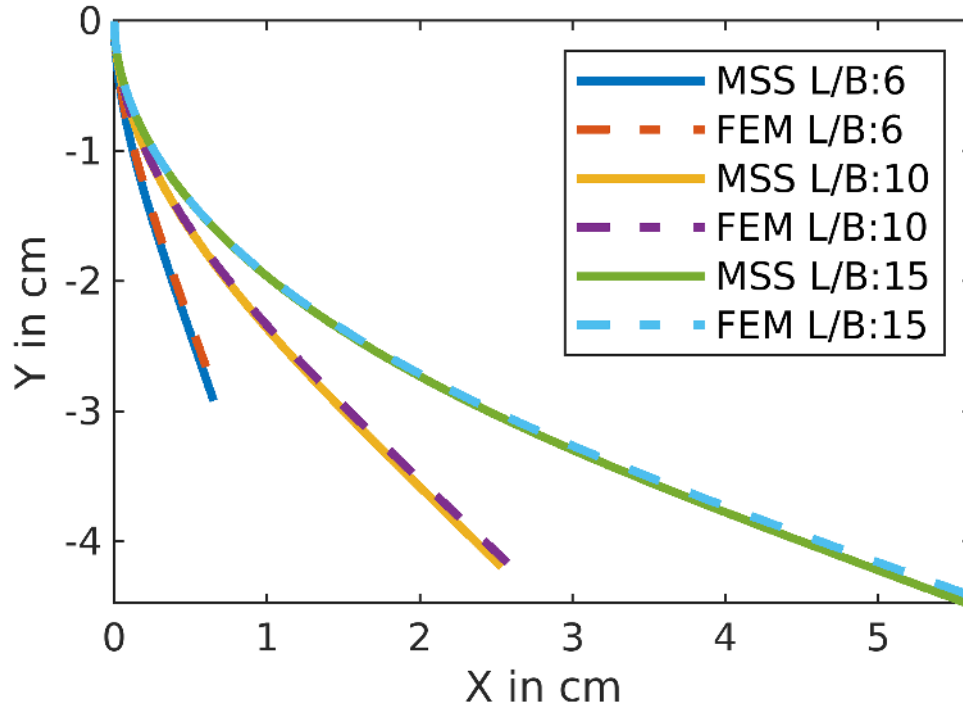


Figure 3.3: Bending of the beam under uniformly distributed bending load. The predictions of the proposed model are compared with FEM results.

With reasonable confidence on static predictions, the dynamic response of the model is compared with beams dynamics response. For this natural frequency of beam are compared with FEM results shown in table. 3.2. The error in natural frequency is within 2% for up-to 3rd natural frequency.

Table 3.2: Natural Frequencies Comparison

Dimensions	Natural Fre- quency	FEM	MSS	% error
$1 \times 1 \times 90$	1 st	0.64452	0.65	0.15
	2 nd	4.0365	4.04	0.15
	3 rd	11.2915	11.31	0.15
$1 \times 1 \times 15$	1 st	23.2144	23.18	0.17
	2 nd	142.4335	142.29	0.10
	3 rd	386.4096	386.34	0.02
$1 \times 1 \times 4$	1 st	319.51	312.76	2.16
	2 nd	1601.77	1,583.70	1.14
	3 rd	2048.88	2,025.40	1.16

A step force was applied to a beam of $0.05 \times 0.05 \times 0.75$ with Young's modulus 1Mpa and Rayleigh damping coefficient for mass and stiffness as 0.1 and 0.3124

respectively. As it can be seen in Fig. 3.4 that the the model predicted rise time and other dynamic features agree well with the FEM results. The final settling value doesn't agree well for y tip deflection but even there the absolute error is in the range of 10^{-3} . Moreover, it has already been observed that the static results match very well even for large deformations.

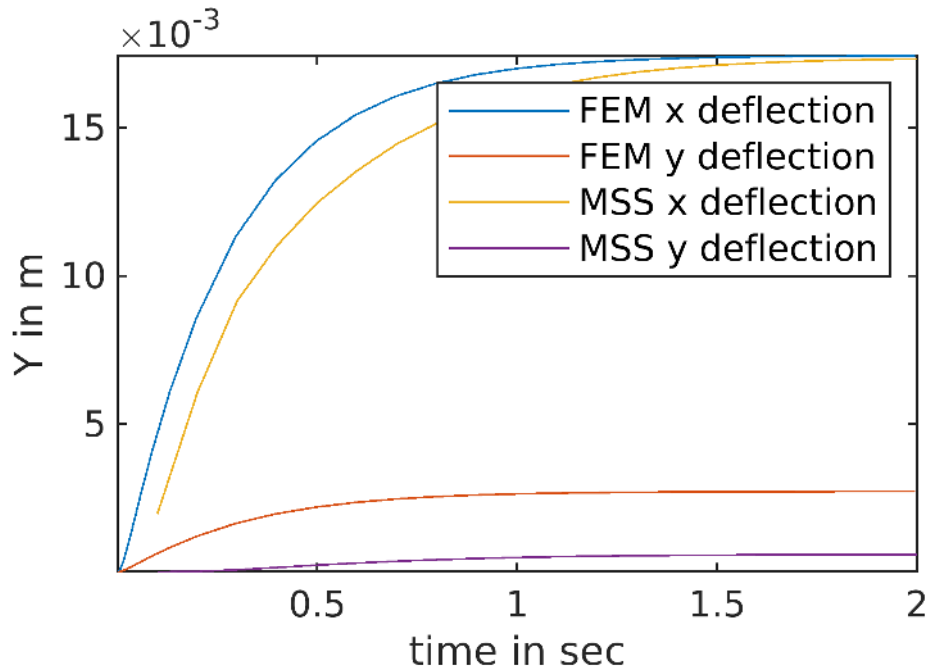


Figure 3.4: The dynamic response of the FEM model compared with the model predictions of a beam under step force applied at the tip at $t = 0\text{sec}$.

Chapter 4

Control Design

Continuum robots are soft mechanical devices, made of deformable soft material with a finite number of inputs, i.e. torque, acting at finite locations along the length of the beam. In the following, we present a controller design for driving a continuum robot to a desired feasible point or making it follow a trajectory. A simplified version of model predictive control is used in this work. As the measurement of complete state, displacements of every node, is difficult we take the measurement at only selected points for feedback. The effectiveness of the controller was tested in simulations hence to introduce model uncertainties different mesh resolution was used for simulating the real system and the controller.

4.1 controller design

The continuum robot is modeled with 5 beams of $0.05m \times 0.05m \times 0.15m$ with Young's modulus $1 \times 10^6 Pa$ each connected in series with a stiffer plate at the joints to apply pure moment at every section as shown in Fig. 4.1. The objective of the controller is to control the tip position of the robot. At every time step a control



Figure 4.1: Continuum robot design

input is calculated using the step response of each of the inputs to each of the outputs. The controller predicts the behavior of the system under given control

action for a single time step and then applies a step on top of that to each of the inputs individually and measures its effect on the output. This helps us predict the effect of each input on the output which is then used to design the control action for the next time-step. The predicted control input for the next step is given by

$$\frac{\begin{bmatrix} \Delta x_1 & \Delta x_2 & \Delta x_3 & \Delta x_4 & \Delta x_5 \\ \Delta y_1 & \Delta y_2 & \Delta y_3 & \Delta y_4 & \Delta y_5 \end{bmatrix}}{stepheight} \Delta U = \begin{bmatrix} Xd_{i+1} - X_i \\ Yd_{i+1} - Y_i \end{bmatrix} \quad (4.1)$$

$$\Delta U = \begin{bmatrix} \tau 1_i - \tau 1_{i-1} \\ \tau 2_i - \tau 2_{i-1} \\ \tau 3_i - \tau 3_{i-1} \\ \tau 4_i - \tau 4_{i-1} \\ \tau 5_i - \tau 5_{i-1} \end{bmatrix}, \Delta x_i = Xsp_i - Xp$$

where, τj_i is the calculated control input for time instance i , Xd_{i+1} is the desired point at time instance $i + 1$, X_i is the measured tip pose at time instance i , Xsp_j is the step response of input j and Xp is the predicted pose at time instance i under the control action at $i - 1$. This control strategy works under pseudo-static conditions as far as the system behavior is linear in a small regime. The δU is calculated from equation 4.1 using the left pseudo-inverse of the step matrix. The max of calculated ΔU is saturated and other entries are scaled in a similar ratio to maintain the direction of new ΔU to be the same as before saturation. The saturation is done because the predicted direction of the force gives movement in the desired direction only in a small linear regime.

This entire framework is implemented using Robot Operating System (ROS). To replicate the behavior of a real system we simulated our model at a higher mesh resolution such that 8 square element is present along the depth of the beam. The control action is predicted using a low-resolution model such that only 3 square elements are present along the depth of the beam. This is to ensure feedback from the actual experimental setting for every node of coarser mesh can be extracted by using visual-inertial state estimators [11]. The prediction should be faster than the real system such that the predictions of pose under step and without step with control action are ready before the next time step. This would ensure that as soon as the measurement is received the next control input can be calculated. Thus, parallel computation of step responses is done using ROS nodes as shown in Fig. 4.2. Also, as the predictor is running at low mesh resolution we can easily simulate the system predictions for as much as 15 time-step in T sec, where T is the time-step size.

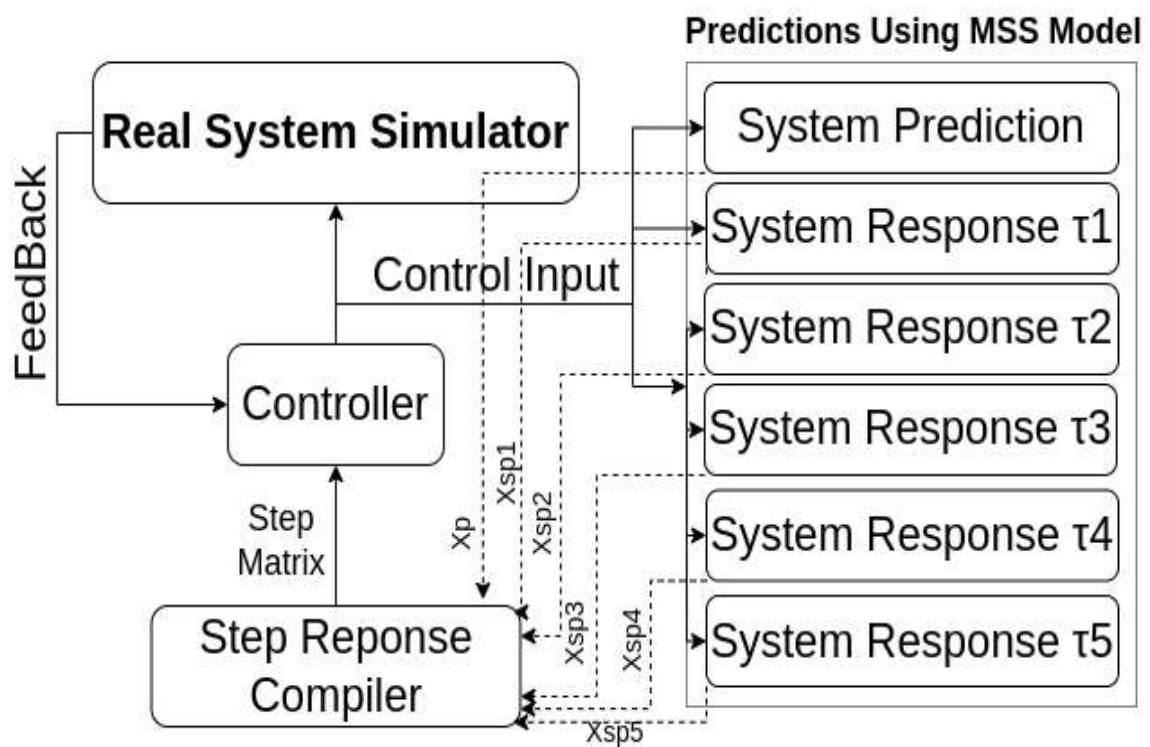


Figure 4.2: ROS framework of the implemented system for simulating the real system and running model predictions in parallel for control action decision along with information flow is pictured.

Simulation Results

The controller was tested for tracking points as well as for following trajectories. When trying to move to a point, the controller could take the tip to desired tip pose as far as the point lied in the workspace of the continuum robot. Fig. 5.1 shows the motion of the robot under the controller action when the robot at rest was asked to move to a certain point. As at the initial position directly moving in the direction of desired position is not possible, the controller takes the robot through a set of configurations such that the favourable direction at that configuration and desired direction to move in both are accounted for. This effect is due to the step matrix in equation(4.1) whose eigen values and the left and right eigen vectors gives the manipulability ellipsoid.

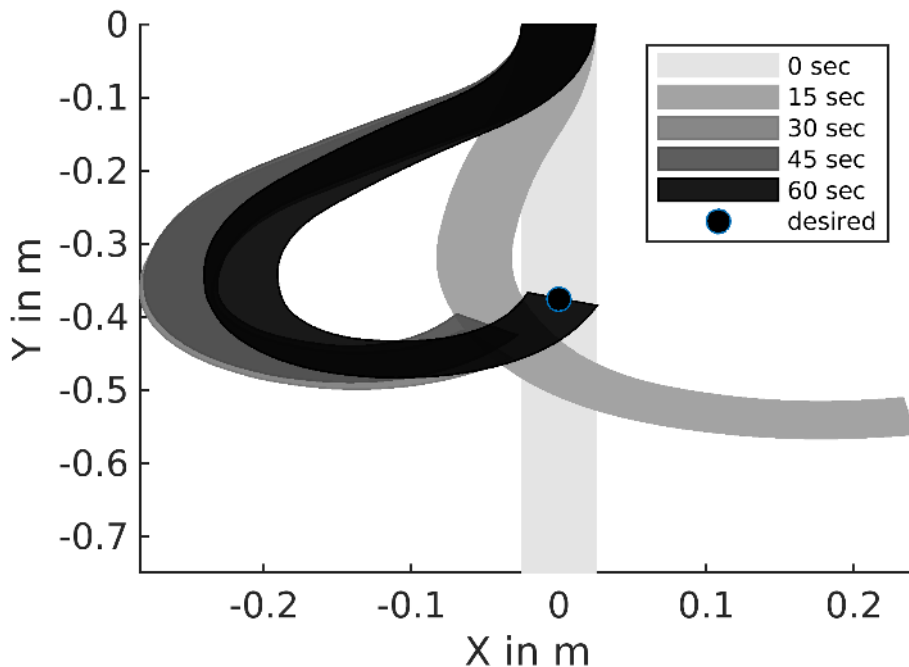


Figure 5.1: going to a point

For following trajectories, the controller was tested for varied trajectories at multiple different places in the workspace of the robot. Fig. 5 shows the controller response to following an ellipse about the center-line of the robot's initial configuration. It

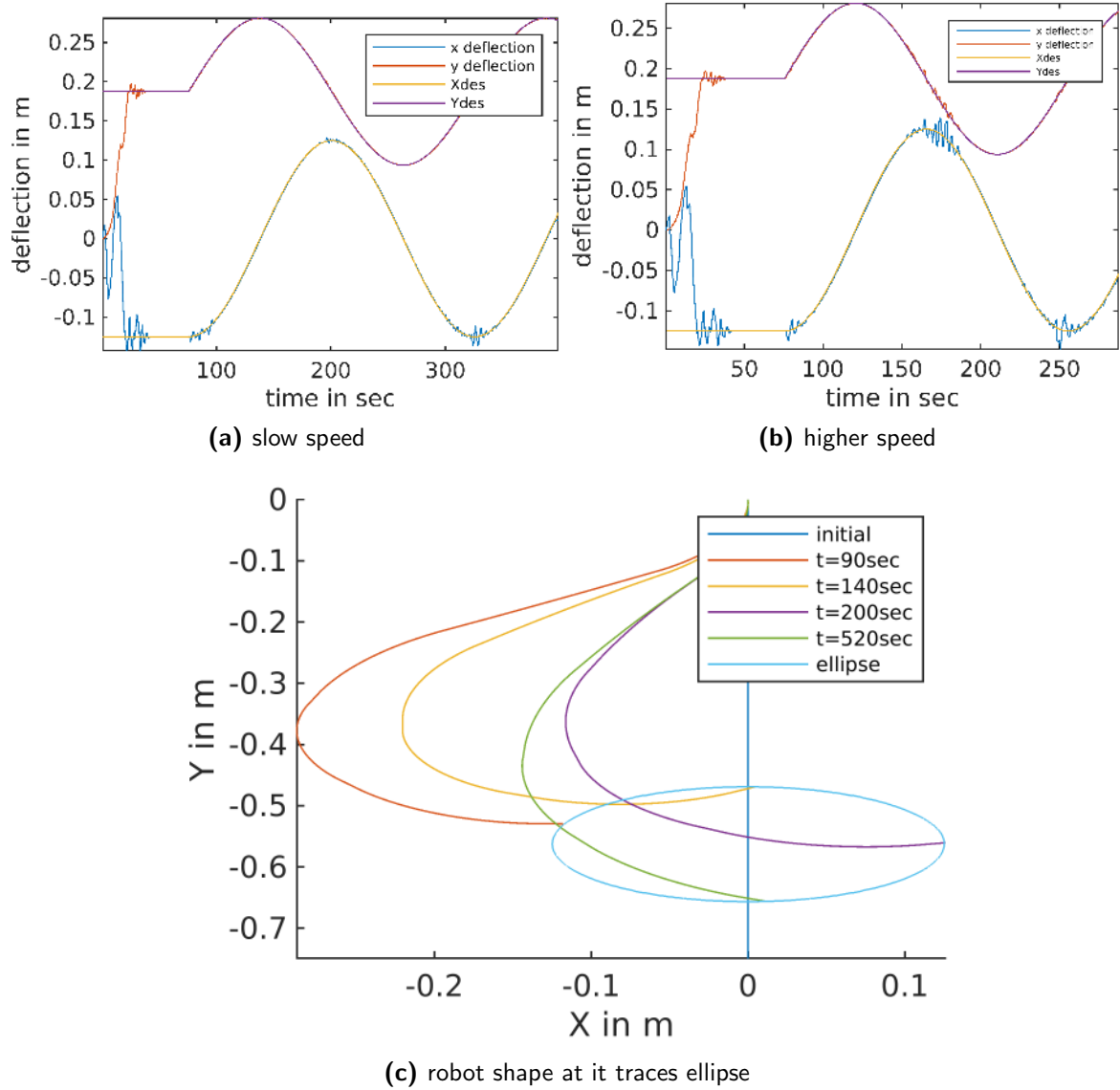


Figure 5.2: Following an elliptical path with the minor axis of the ellipse along the center-line at the initial configuration of the robot.

can be seen that the trajectory is followed relatively well along most of the path except when the direction changes sharply near the tip of the major axis. At higher speeds, it is difficult to keep the robot tip along the trajectory while a change of direction at the end of the major axis of the ellipse. The initial oscillation while trying to move to trajectory from the rest position can be attributed to the transient high-frequency component of the robot.

Hardware Design and Fabrication

The continuum robot consists of 5 independent segments joined together in series as shown in Fig. 6.1. Strings are used to control the torque being applied at the tip of each segment.

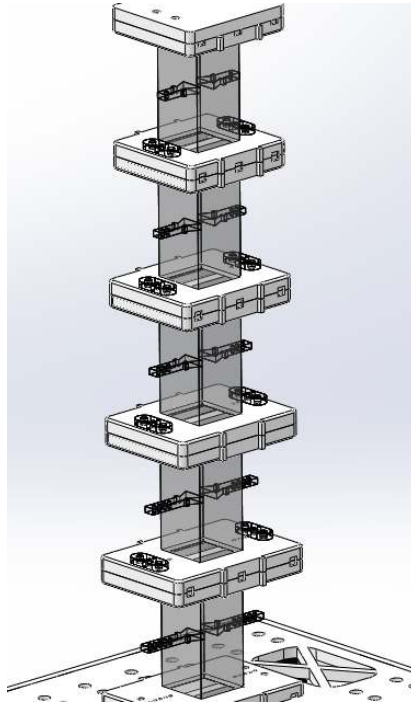


Figure 6.1: Design of the 5 segment 2D continuum robot

6.1 Fabrication of single segment

Each segment is made up of a soft central core with a thin rigid link attached at either end of the core as shown in Fig. 6.2. The rigid ends are used to apply force to the segment and also to attach it to other segments. A Servo motor [30] is used to control the position/force being applied on the string. The fabrication of the segment is done by casting as shown in Fig 6.3. The rigid link is placed in the mold of the required shape and PDMS is then poured.

Due to the high gear ratio within the servo, it was not possible to control the torque being applied on the segment by commanding the PWM to the motor. However, the slope could be controlled by commanding the position of the servo. As the controller design required commanding torque an attempt was made to design a torsion-based torque sensor using the same soft material PDMS that was used to fabricate the segments.

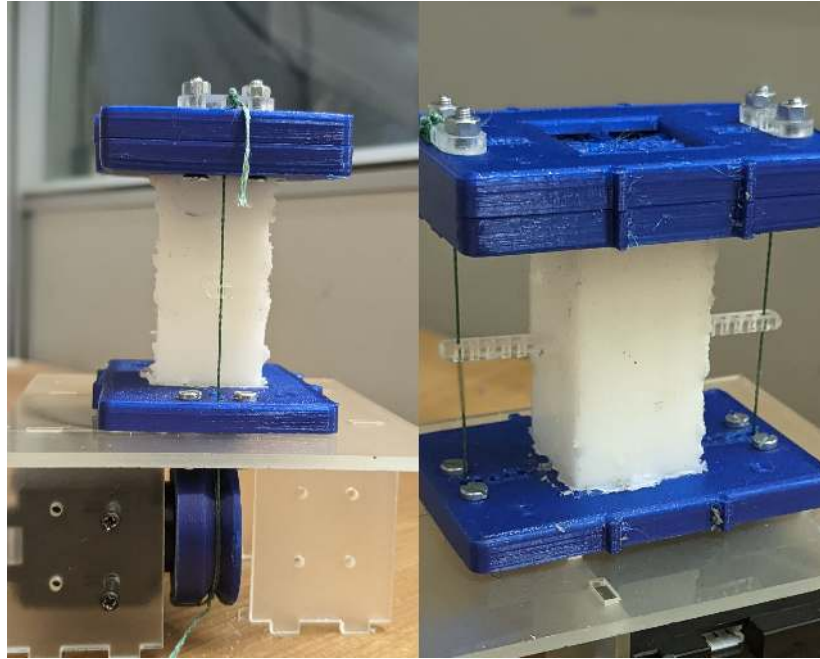


Figure 6.2: One soft segment with the central core made of silicone polymer. The blue parts are 3D printed with ABS and provide the necessary rigidity to apply a distributed torque at the tip of the segment. The strings connected to the servo through the pulley act as an actuator.

6.2 Torque sensor Design

A simple torsion-based sensor was made using the same fabrication technique as above Fig. 6.4. The torsional stiffness of the sensor could be changed by changing the length and diameter of the sensor. Based on the max shear stress-induced and the deformation, the length and diameter could be optimized to have good range and resolution. Sensors of different lengths and diameters were made and tested. To characterize the torque versus theta response of the sensor, a string was wound around a pulley attached to the end of the sensor, and known weights were hung from it as shown in Fig. 6.5. A linear behavior was observed up to a range of $\pm \frac{2\pi}{3}$ between torque and theta and the same was used to characterize the torsional stiffness of the sensor. The sensor being made of soft material its property changed over time and its fatigue life was very poor.

After characterization, the sensor was mounted between the servo and an encoder as shown in Fig 6.6. A pulley was attached to the sensor on the encoder side and the string wound around the pulley actuated the segment. When the sensor was tested with one segment, it was found that the range of torque that the sensor could transfer was very limited. When trying to optimize the dimensions of the sensor

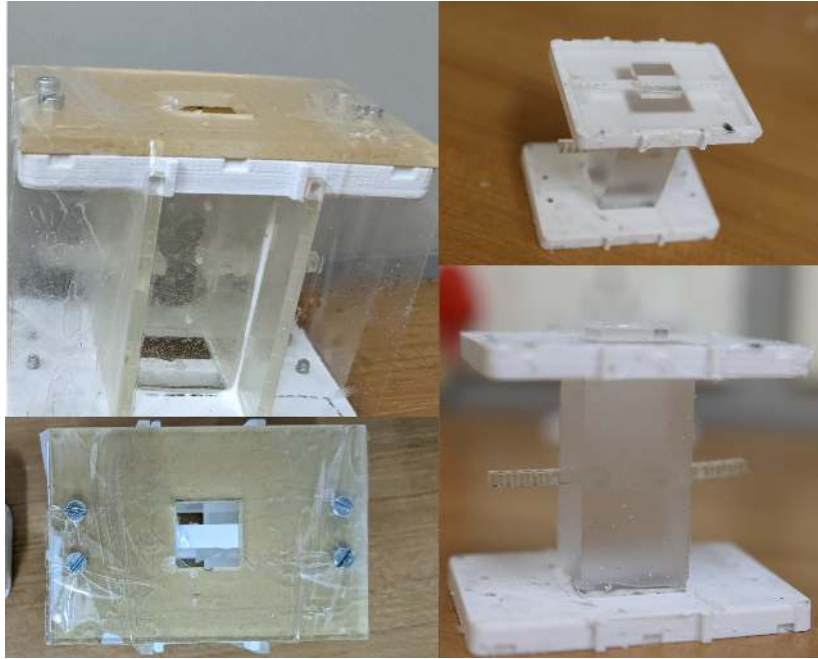


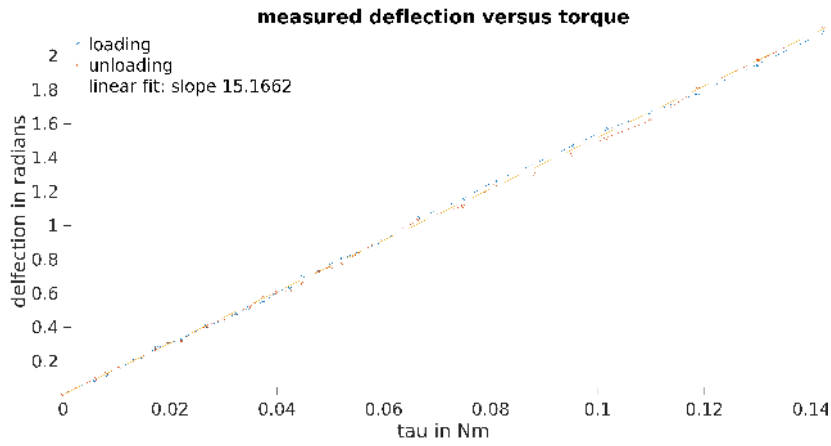
Figure 6.3: The images on the left show the mold, where the rigid attachments and string guide have been put in the mold. After pouring and curing the mold is removed and the extra material on the top is cut.



Figure 6.4: The mold for the cylindrical torque sensor is 3D printed in 2 parts. On either side, a rigid part is put in to transfer the load and act as support for bolting it to other parts.



(a) Experimental setup design



(b) theta vs torque graph

Figure 6.5: For sensor characterization, a mass of known weights is hung from the string, and the deformation is recorded through the encoder while the servo is commanded to hold the position.

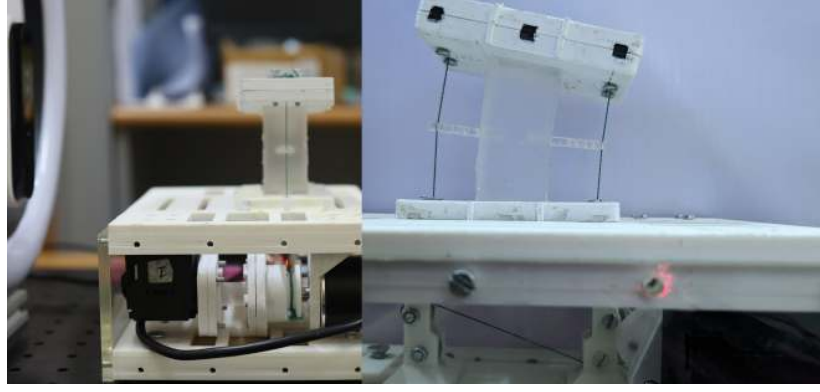
to increase the range, the resolution would be sacrificed which would then be the bottleneck in achieving high resolution in continuum robot movement. Furthermore, it was observed that the relation between the slope of the segment and the torque was linear in the range we tested the sensor. Hence, it was decided that the controller could be modified to command slope instead of torque based on this linear relation.

6.3 Assembly of full robot

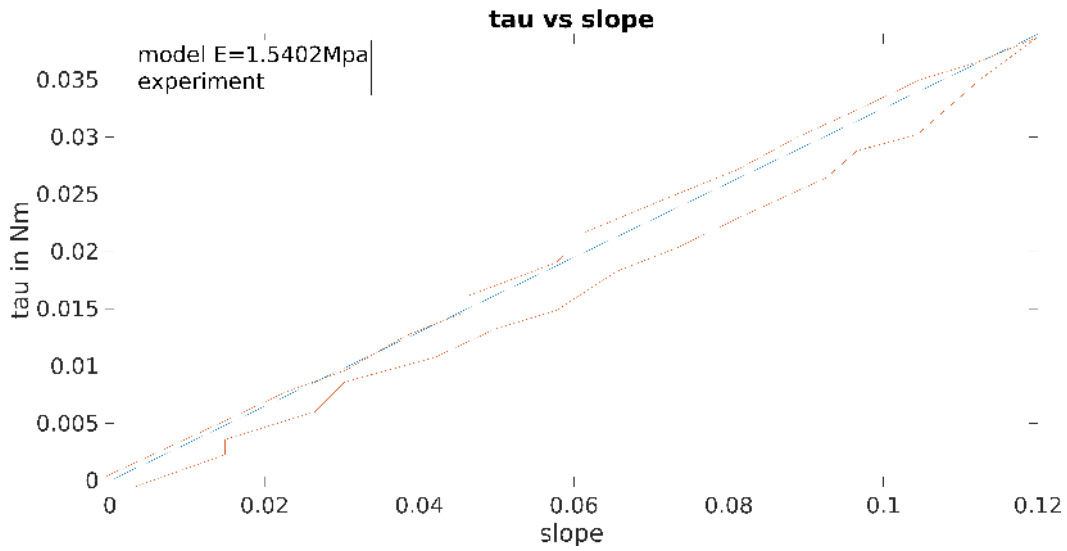
A final version of the continuum robot was made by combining 5 segments, each actuated with a different string connected to a servo as shown in Fig 6.7. The continuum robot is then mounted on a sica frame and placed on a horizontal plane.

6.4 Sensing

A 4K cam mounted on the Sica frame is used for sensing. Three markers arranged in a line at an equal distance are placed on each segment. The image coordinate of each dot is extracted using an image processing technique which is then converted from pixel coordinates to Cartesian coordinates using a fish-eye-lens model [29]. The position and slope of each segment are estimated using the position of the



(a) experimental setup design



(b) slope versus torque graph

Figure 6.6: The torque applied on the segment is transferred through the torque sensor. The slope of the segment for any given torque is captured through the camera.

three dots on the segment. The sensing resolution and other characteristics are represented in Fig. 6.8. The error in pose measurement is within 0.35mm, while the error in slope measurement is within 0.005 radians.

6.5 Full Control Architecture

With difficulty in achieving torque control with existing hardware, it was decided that the high-level controller should be modified to command slopes instead of torque. This was possible because the slope vs torque relation was almost linear in the small regime and quadratic otherwise as shown in Fig. 6.9. Hence, estimating torque from the slope and vice-versa, helped in making necessary modifications to the controller. Along with the proposed controller at the high level, a low-level PID-based slope controller was developed. The modified control architecture is shown in Fig. 6.10. At every time-step, the high-level controller gives out the desired torque on every section which is then converted into the desired slopes at each section using the pre-tuned scaling factor. The calculated slope is then commanded to the slope control which commands the servo position. The measurement of



Figure 6.7: Left images shows the complete assembly of the 5-segment continuum robot with markers placed on each segment for tracking. The top right image depicts the robot being placed on a horizontal frame with the camera mounted on the top, while the bottom right image depicts the 5 servos being used for actuating the segments.

slope at every time step is used to estimate the torque that would be acting on every segment. This calculated torque along with the output states is passed to the high-level controller. The high-level controller simulates the system for the given torque to bring the model to a state close to the actual hardware state, after which it calculates the step and then estimates the new torque based on desired and present output states.

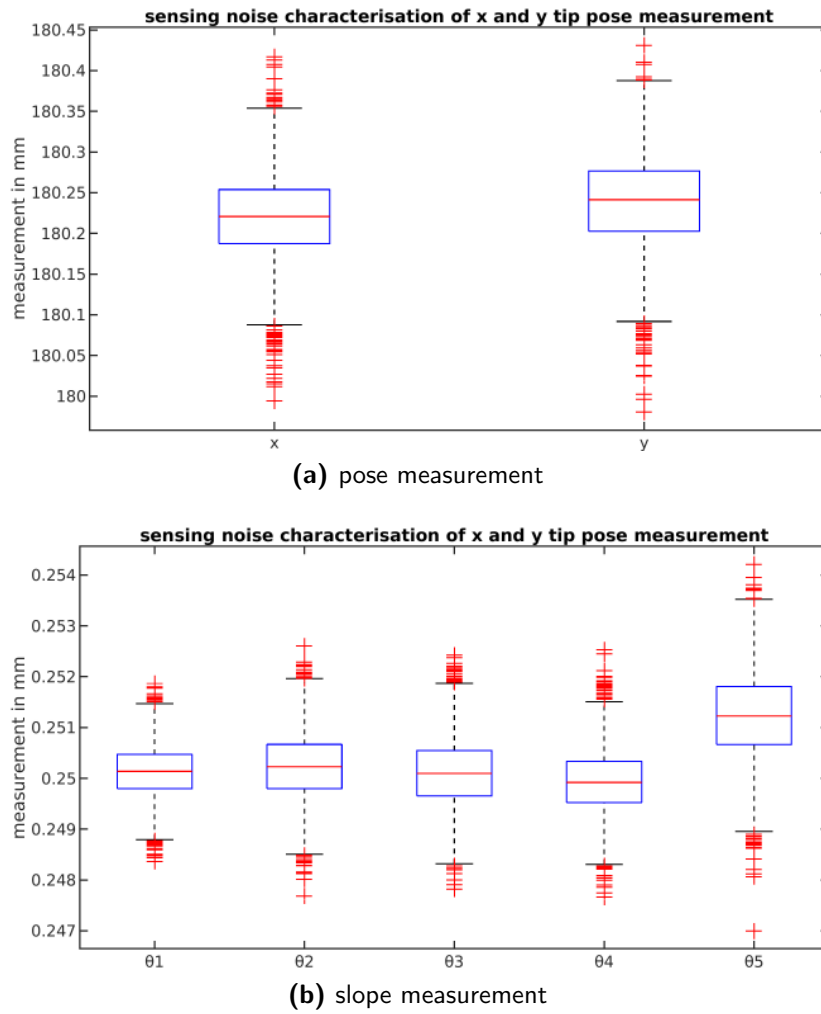


Figure 6.8: Box plot of measurement of pose and slope at a given position is shown in the above plots.

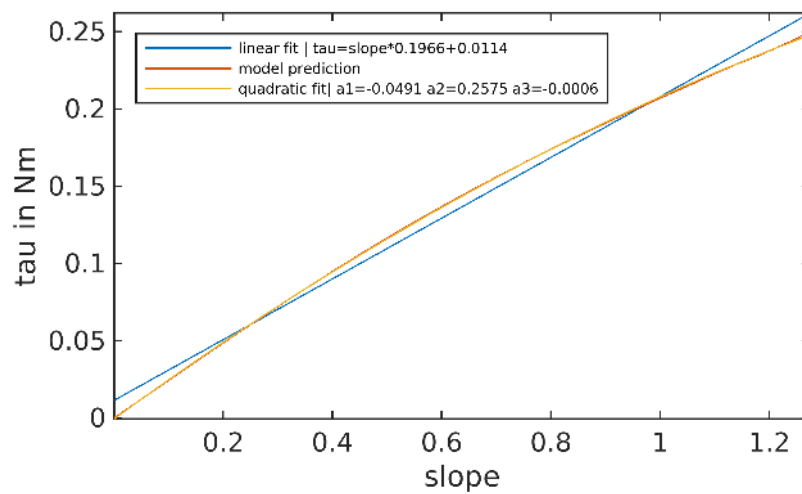


Figure 6.9: variation of required torque with increasing slope predicted from the model is shown here.

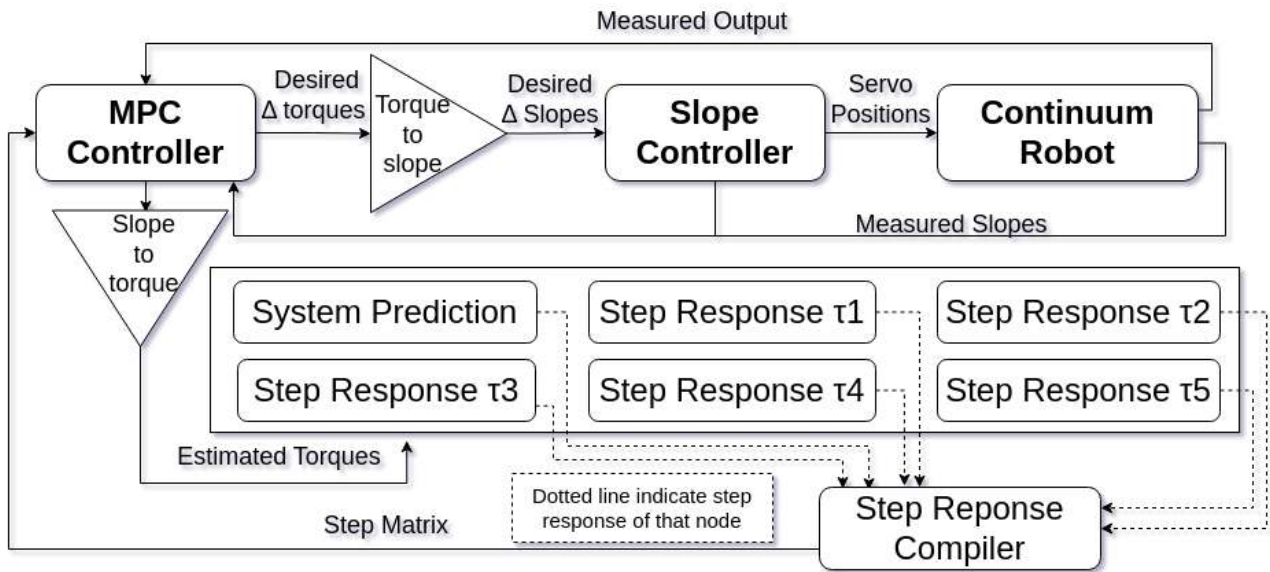


Figure 6.10: The control diagram of the modified controller after implementing slope control at a low level as per hardware requirement.

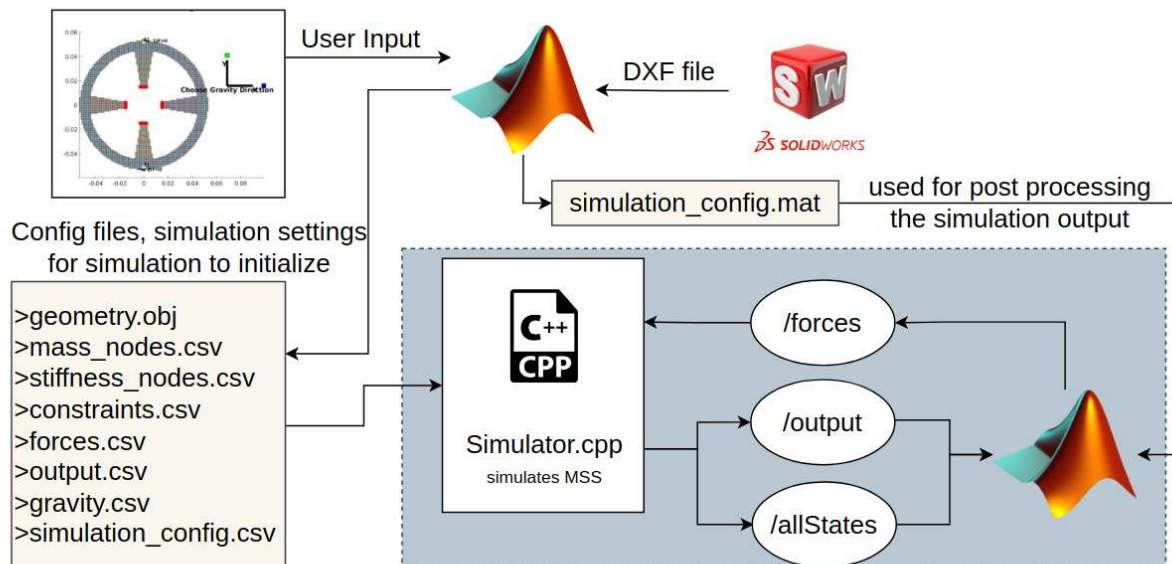


Figure 6.11: The flow diagram explaining the hardware architecture of the experimental setup

Experimental Results

Experiments were conducted on the 5-segment continuum robot. The results of the slope controller, tip pose set-point control and tip pose trajectory control is discussed in this section.

7.1 Slope control Results

The low-level slope control is based on PID. When moving to a desired slope the controller shows an accuracy of up to 0.005 radians, limited by the sensing accuracy. The rise time of the system is around 1s. The bandwidth of the controller is around 0.25Hz, the slow speed is expected because the controller is a pseudo-static controller and doesn't account for the dynamics of the system. The results of the slope control are shown in Fig. 7.1. It can be seen that there is a slight lag in slope control tracking.

7.2 Set-point Control

When trying to move the tip of the robot to a desired position, the controller works with an accuracy of 0.3mm. The Controller takes the continuum robot to the desired position if the pose is within the reachable workspace. The reachable workspace of the robot depends on the length of the segment and the maximum allowed relative slope between 2 segments. The control results are shown in Fig. 7.2

7.3 Trajectory tracking

The controller in discussion is pseudo-static, hence the trajectory tracking performance was found to be good for a slow-speed trajectory with the average error being less than 1mm. At a relatively higher speed, the tracking accuracy decreases with increasing speed. As shown in Fig. 7.3, the tracking is very good in the constant velocity regime, but as the path changes its direction, if the change in direction is sharp the tracking performance decreases a little and it then picks up quickly and starts tracking within 0.3mm.

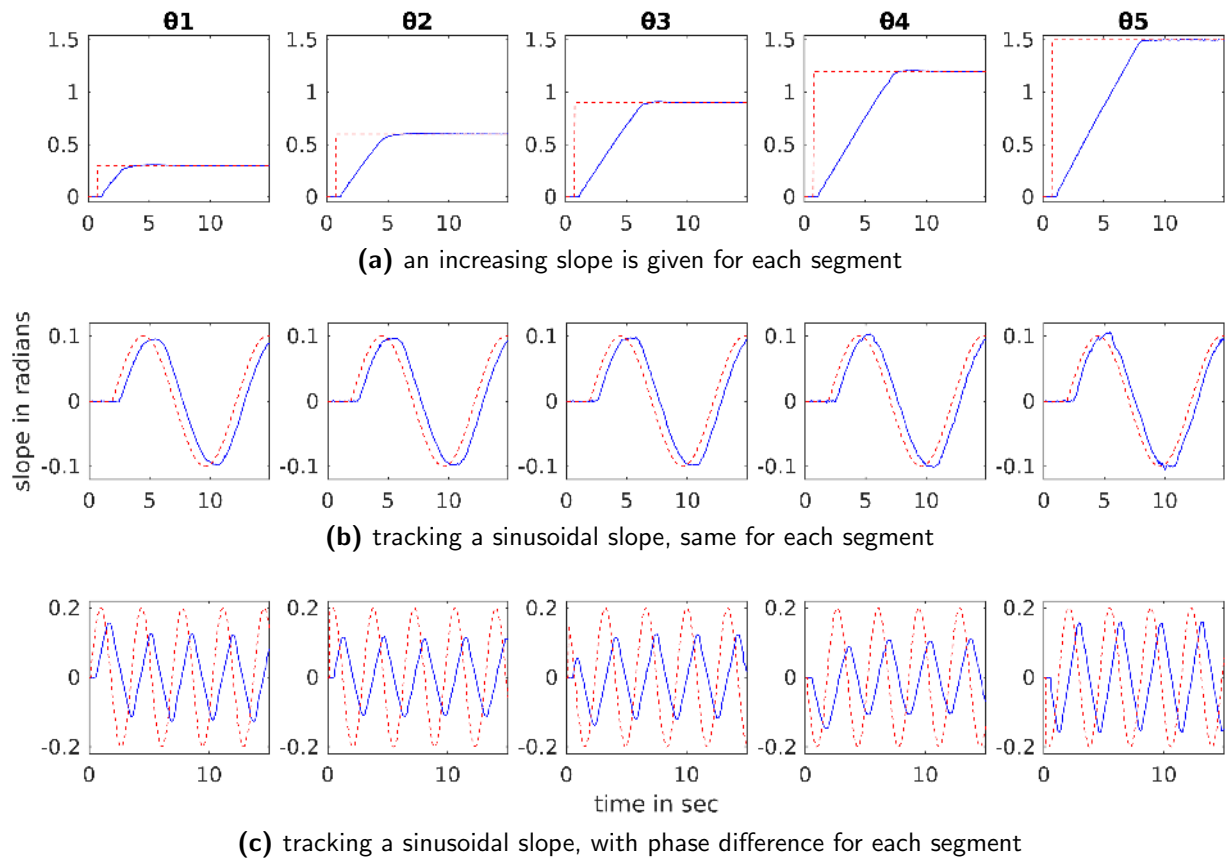


Figure 7.1: Slope versus time plot obtained while tracking absolute slope as specified by the red dotted line for each segment.

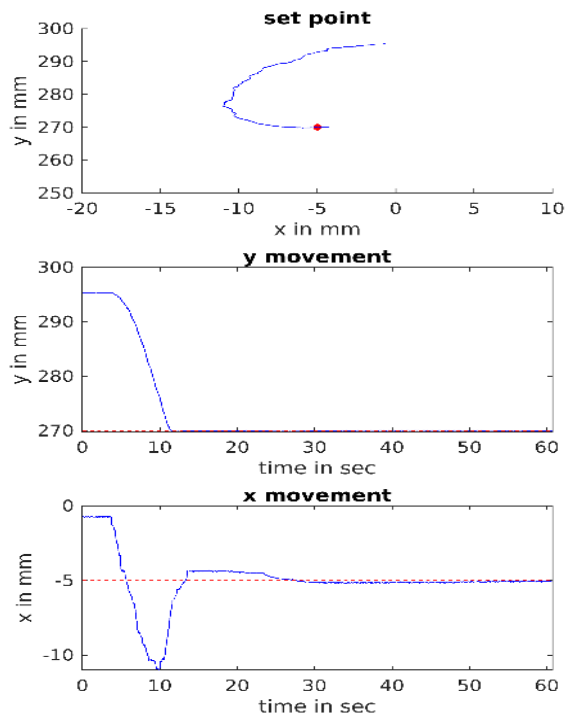
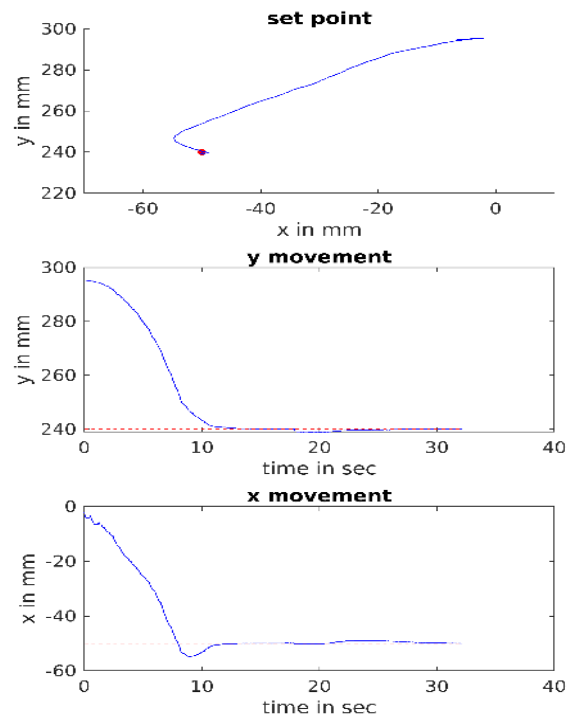
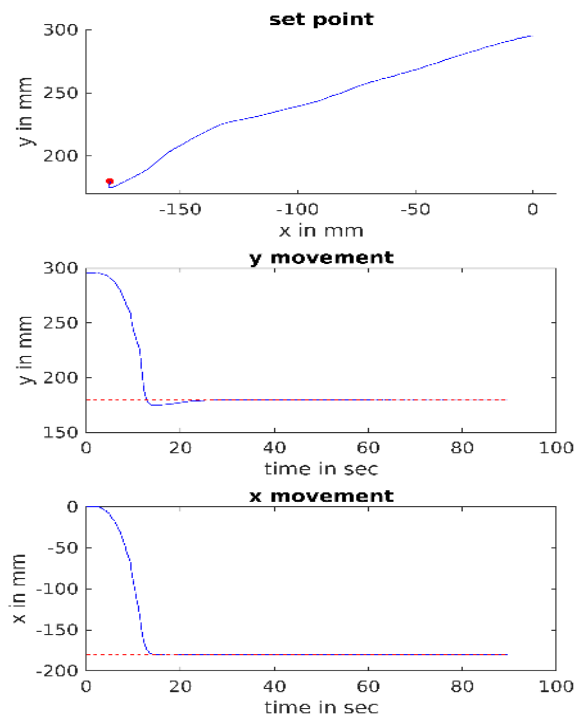
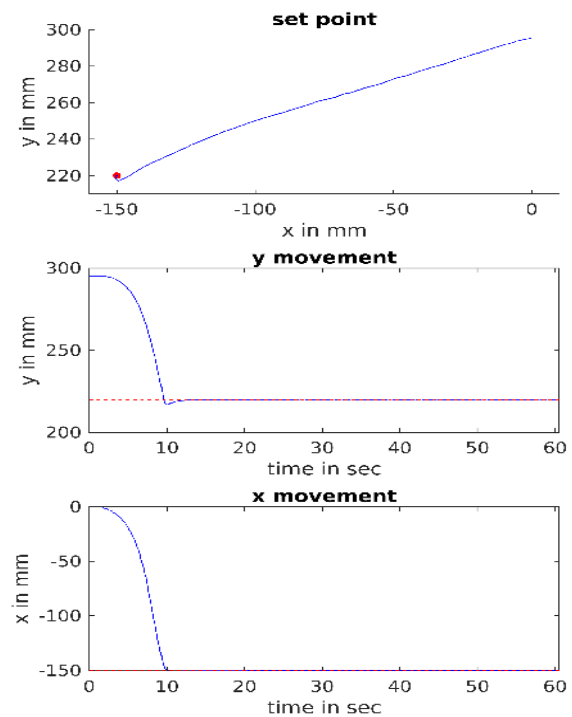
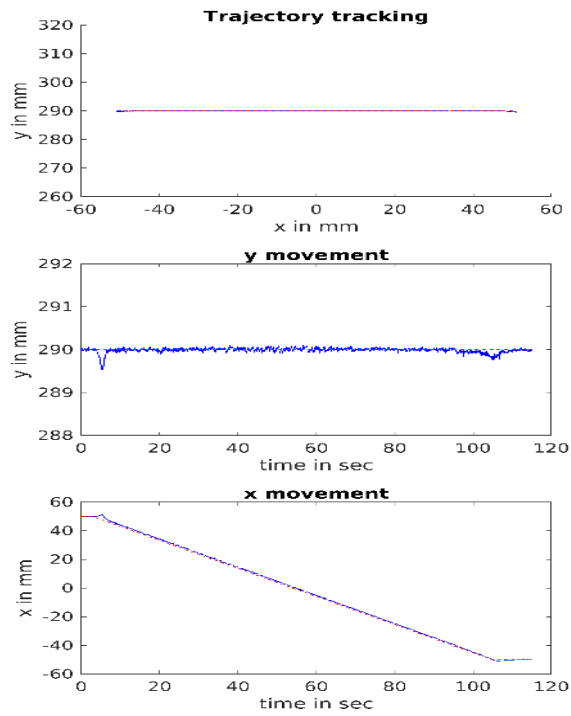
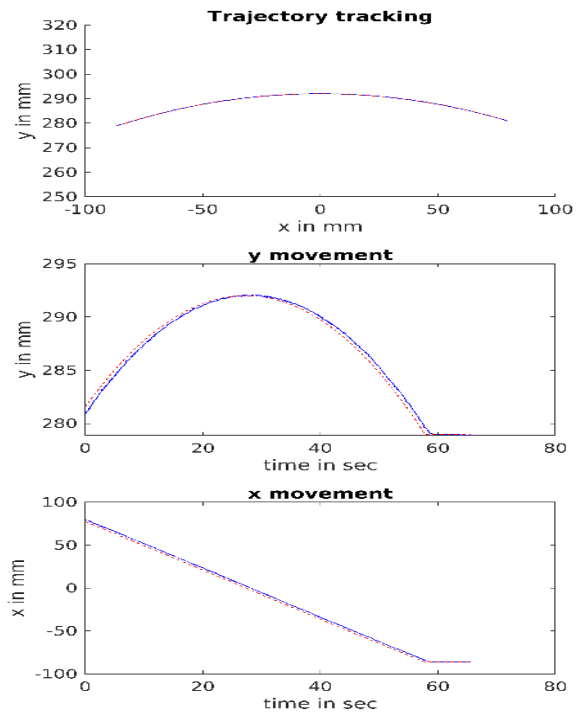

 (a) set point: $(-5, 270)$

 (b) set point: $(-50, 240)$

 (c) set point: $(-180, 180)$

 (d) set point: $(-150, 220)$

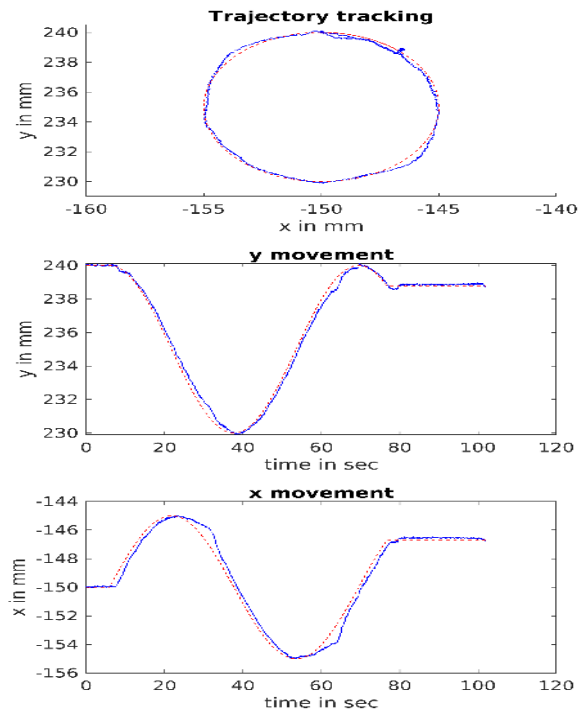
Figure 7.2: The path and time taken to move from zero-config to various set-point using the proposed controller is highlighted.



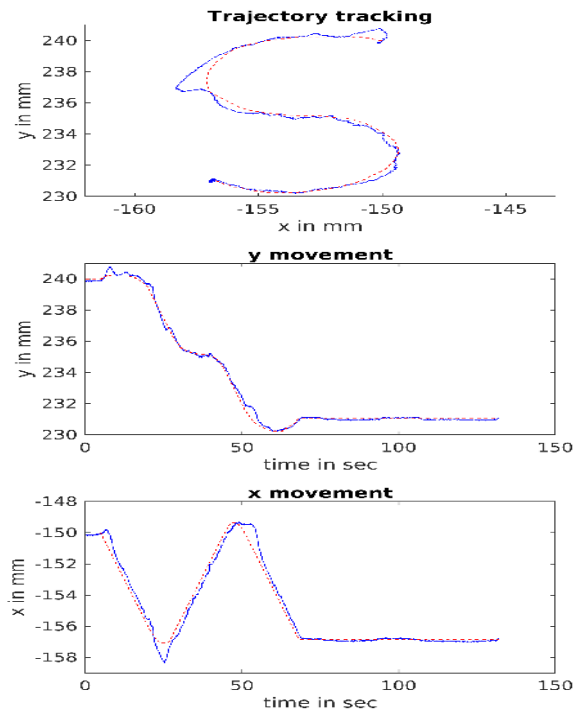
(a) Avg. error:0.8mm



(b) Avg error:2.2mm



(c) Avg. error:0.6mm



(d) avg error:0.6mm

Figure 7.3: The performance of the controller while following different trajectories in the workspace is depicted.

Discussions and conclusions

The proposed model for the prediction of the behavior of soft robots was benchmarked with existing models and FEM. The model performed well in all of the bench-marking with the error being within 6%. When the proposed control strategy was tested on a 5-segment continuum robot the control performance was great for slow trajectories. The controller struggled in cases where the velocity changed sharply but the good thing was the controller could catch up with the trajectory eventually. The observed behavior is expected owing to the pseudo-static nature of the controller. Even then the performance of the controller is much better than any other existing controller for soft robots.

The workspace of the continuum robot is dependent on the maximum allowed relative slopes between the two segments. The knowledge of maneuverability at any given position and the overall workspace of the robot can be used to perform dynamic planning of trajectories which could result in much better tracking performance and the capability of using attaining a larger task space.

With the ability to quickly simulate different 2D soft robots and ease of modeling algorithmically, this work can be used to come up with innovative 2D soft robot designs through various optimization technics.

Using model-order reduction or exploring PDE-based control is another domain in which this work can be extended. With this, a better and more robust control strategy could be developed which would account for the dynamics of the robot and could result in the tracking of faster trajectories.

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